VASU GUPTA
22V3062

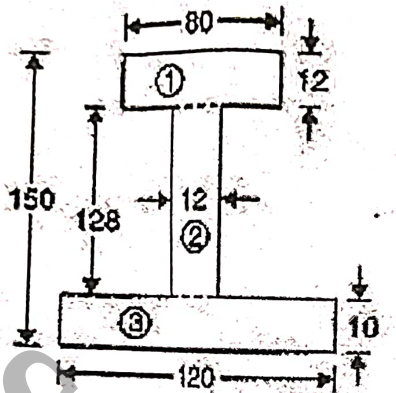
BE II EXAMINATION DEC 2023
CIVIL ENGINEERING
3VLRC2
APPLIED MECHANICS & STRENGTH OF MATERIALS

2024046

Duration: 3 Hr **Max. Marks: 60s**

Note: **Attempt any two parts from each question.** All questions carry **equal marks.** Assume suitable data wherever required. **All parts of the same question must be solved in continuation.**

Q1. (a) Find the moment of inertia of an I section as shown in figure about X-X axes through the centre of gravity of the section. All dimensions are in mm. 12.47×10^6 mm

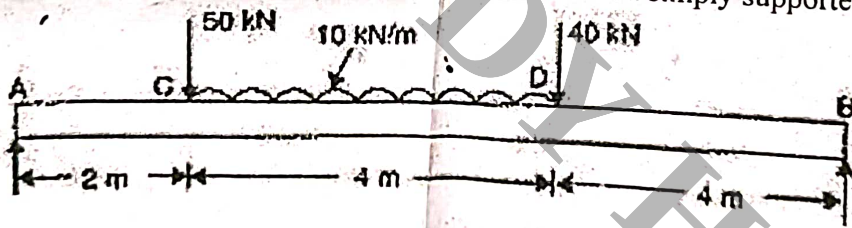


(b) Derive an expression for moment of inertia of a rectangular lamina of width 'b' and depth 'h'.

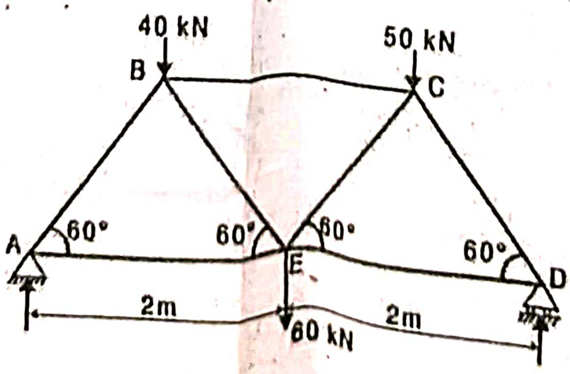
(c) Explain the various theorems of moment of inertia.

Q2. (a) *By using the principle of virtual work,* find the reactions of a simply supported beam.

50 = RB
80 = RA



(b) Find the forces in the members of truss as shown in fig.



(c) Write a short note on **any one.**

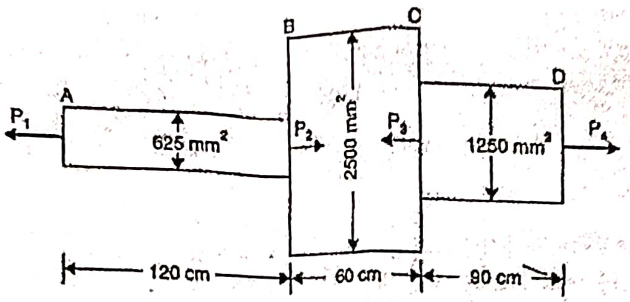
i. Types of trusses.

ii. Principle of virtual work.

P.T.O

Q 3. (a) A round bar, of length L, tapers uniformly from small diameter d_1 at one end to bigger diameter d_2 at the other end is subjected to a tensile axial load P. Derive an expression for the extension produced

(b) A member ABCD is subjected to point loads P_1, P_2, P_3 and P_4 as shown in figure. Calculate the force P_2 necessary for equilibrium, if $P_1 = 45$ kN , $P_3 = 450$ kN and $P_4 = 130$ kN. Determine the total elongation of the member. Take $E = 2.1 \times 10^5$ N/mm^2

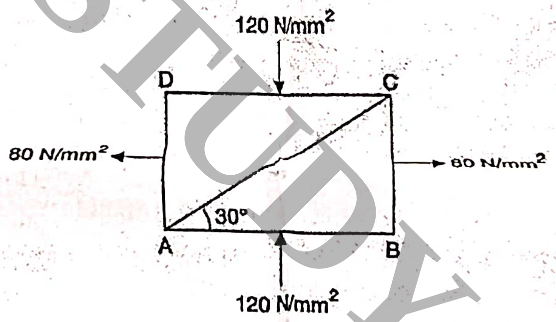


0.4914mm = dL

(c) write a short note on (any one)

i. Elastic constants
ii. Mechanical properties of materials

Q 4. (a) The direct stresses at a point in the strained material are 120 N/mm^2 compressive and 80 N/mm^2 tensile as shown in Fig. Find the normal and tangential stresses and resultant stress on the plane AC.



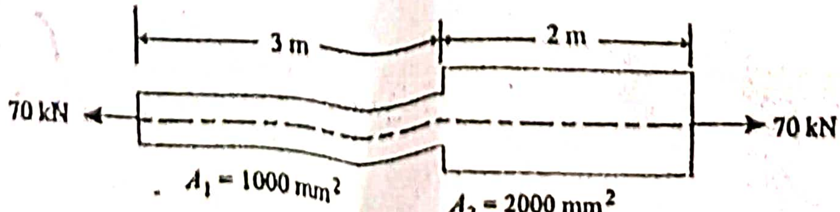
(b) Explain Mohr's circle method for determining normal, shear and resultant stresses on an oblique plane subjected to two mutually perpendicular direct stresses.

(c) Derive an expression for Normal and Tangential stresses on an inclined plane subjected to two mutually perpendicular direct stresses.

Q 5. (a) A non-uniform tension 5 m long is made up of two parts as shown in fig below. Find the total strain energy stored in the bar, when it is subjected to a gradual load of 70 kN. Also find the total strain energy stored in the bar, when the bar is made of uniform cross section of the same volume under the same load. Take $E = 2 \times 10^5$ N/mm^2.

$U_A = 49000$

$U_B = 43750$



(b) State Castigliano's First Theorem. Using it, find the deflection of simply supported beam carrying a point load P eccentrically on the span. Assume uniform flexural rigidity.

(c) State and prove Maxwell Reciprocal Theorem.

$y_{AB} = y_{BA}$